

The Open Anchoring Quest

Tackling the reliability problem and boosting the power of anchoring research



Anchoring

Is the current temperature in Ulm above or below 35°C?

- ☐ above
- ☐ below

What is the current temperature in Ulm?

Reliability Problem of Anchoring

- Anchoring scores are usually not very reliable
(e.g., $\alpha = 0.11$, [Röseler et al., 2019](#))
- This did not keep researchers from analyzing correlations between susceptibility to anchoring and personality traits for more than a decade
(e.g., [McElroy & Dowd, 2007](#); [Eroglu & Croxton, 2010](#); [Furnham et al., 2012](#); [Perry & Sibley, 2013](#); [Jasper & Ortner, 2014](#); [Caputo, 2014](#); [Cheek & Norem, 2019](#); [Röseler et al., 2019](#))

Average Correlations with Traits ([Röseler, 2021](#))

Moderator	<i>r</i>	CI 95%	<i>k</i>	<i>N</i>
Neuroticism	.00	[-.08, .08]	6	1911
Extraversion	-.06	[-.13, .02]	7	2479
Openness	.04	[-.05, .13]	8	2549
Agreeableness	.03	[-.08, .14]	6	1909
Conscientiousness	.06	[-.02, .14]	6	1910
Cognitive Ability	.00	[-.02, .02]	5	1165
Cognitive Reflection	-.01	[-.08, .06]	10	1242
Self-Control	-.02	[-.08, .05]	3	875

Note. *r* = Bravais-Pearson correlation, *CI* = confidence interval, *k* = number of studies, *N* = sample size; Effect sizes are REML estimates and were not corrected for publication bias; heterogeneity was low to medium.

- Meta-analytical correlations are extremely close to zero.
- It is possible that susceptibility to anchoring is not a personality trait ([Hedge et al., 2018](#))
- You cannot say “there is no black swan” after having looked only at a handful instances of swans

Possible Origins of the Reliability Problem

- How to find a black swan?
- Anchoring research is heterogeneous with respect to...
 - Anchoring Items
 - Anchors (distances from the true or unanchored value)
 - Comparative questions
 - Adjustment direction (known vs. unknown)
 - Analyses
 - Computation of Susceptibility Scores
 - Participants
 - ...
- There is little theoretical guidance
- It is quite costly to conduct a study testing all these possibilities



OpAQ: Unus pro omnibus

Studies	Sample.size	Items	Trials
7	1756	104	13529

We are accumulating trial-level data for all open anchoring datasets and are reaching out to authors for their original data

This is what this looks like



participant_id	sex	age	anchoring_item	true_value	anchor	anchorhigh	estimate	tasktype	direction	incentive	comparative_question	experiment_type	stimulitype
169	1	19	Height of Mount Everest	8848	600	0	8150	height	0	0	1	1	1
172	2	22	Height of Mount Everest	8848	14000	1	11600	height	0	0	1	1	1
174	2	20	Height of Mount Everest	8848	14000	1	5500	height	0	0	1	1	1
177	2	21	Height of Mount Everest	8848	600	0	7603	height	0	0	1	1	1
180	2	20	Height of Mount Everest	8848	600	0	800	height	0	0	1	1	1

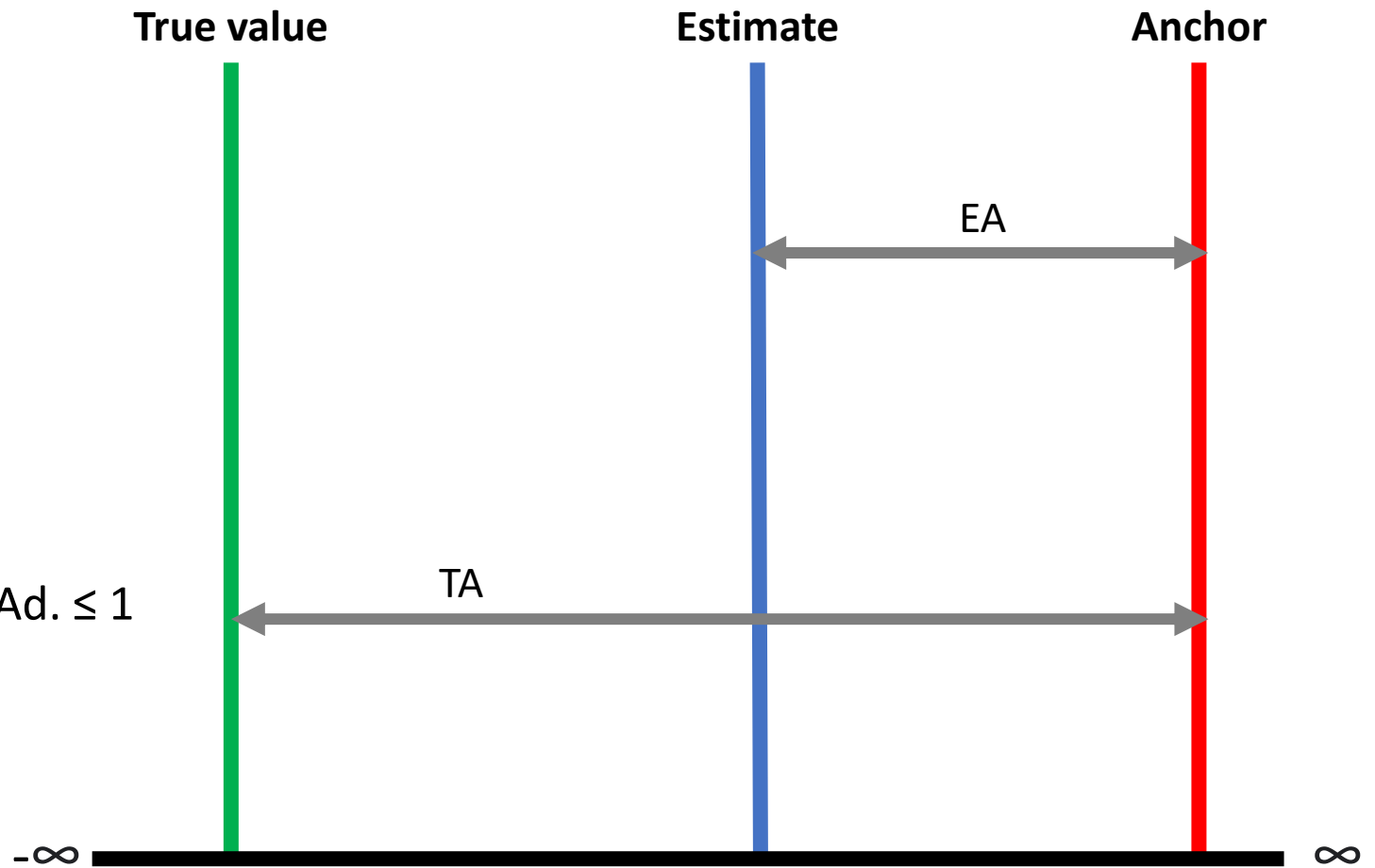
Reliability Scores

Adjustment = EA

Absolute Adjustment = $|EA|$

Relative Adjustment = EA/TA

Relative Adjustment (restricted) = $0 \leq \text{Rel. Ad.} \leq 1$



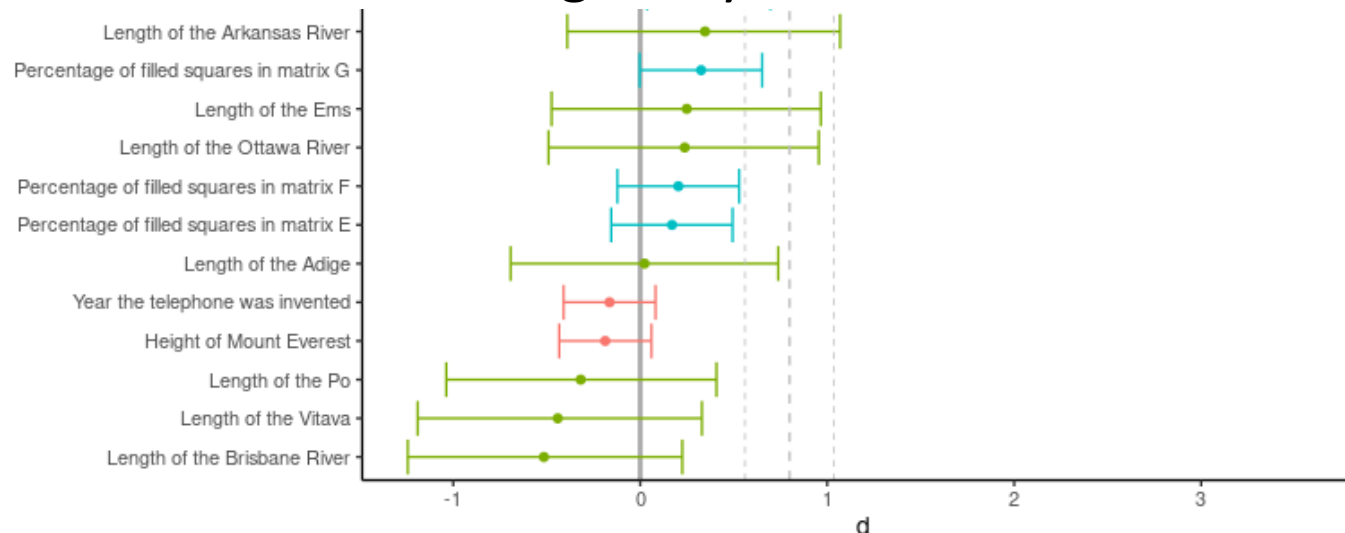
OpAQ Results: Average interitem correlations

reference	items	n	estimate	adjustment	absadjustment	score	restr_score
Bickenback et al., 2021	4	255	-0.082	-0.114	-0.1	-0.145	-0.166
Röseler & Seida, 2021, Pretest	30	30	0.062	0.029	0.212	0.219	0.35
Röseler, 2016	8	145	0.149	0.854	-0.083	-0.038	-0.056
Röseler, Schütz, & Starker, 2019	20	223	0.049	0.049	0.011	-0.015	-0.006
Röseler, Schütz, Baumeister, & Starker, 2020	2	640	0.066	0.069	0.119	0.138	0.21
Yoon, Scopelliti, & Morewedge, 2021, Subscale A	11	97	-0.008	-0.008	0.001	0.045	0.066
Yoon, Scopelliti, & Morewedge, 2021, Subscale B	13	96	-0.012	-0.011	0.054	0.056	0.079
Yoon, Scopelliti, & Morewedge, 2021, Subscale C	14	93	-0.015	-0.003	0.065	0.038	0.085

- Susceptibility to anchoring *can* be measured reliably
- Which score is reliable depends on the study parameters

Potential of OpAQ

- Test hypotheses that are difficult to test in a single study
 - Average effect sizes (Meta-Analysis); currently $d = 0.799$, 95% CI [0.560, 1.037]
 - Impact of item choice
 - Impact of anchor extremeness
 - Effect of demographics
 - Publication bias
- Include non-published datasets that would otherwise stay in the file-drawer
- Become the largest up-to-date anchoring dataset that you want to consult when planning to conduct an anchoring study



The OpAQ-App

t1p.de/openanchoring

OpAQ: Open Anchoring Quest

Info

Summary and Dataset

Anchoring Effect Sizes

Reliabilities

About and References

Welcome to the Open Anchoring Quest!

What is OpAQ?

Research on anchoring effects is currently challenged by the lack of reliability of anchoring scores. That is, susceptibility to anchoring for one item is almost completely uncorrelated with susceptibility to anchoring on any other item. With OpAQ, we try to tackle this problem by meta-analytically investigating different measures of anchoring susceptibility as well as factors influencing their reliability. For this purpose, we have created an open, free, and ever-growing dataset already containing thousands of trials from several studies on anchoring effects – and with your support soon even more.

What are your benefits of joining us?

You are very welcome to contribute data from your anchoring experiment(s)! In return, (apart from being rewarded by the good feeling of helping research on anchoring effects to improve) we will list you as a co-author of the OpAQ. If you provide us your data, we can convert it and add it to the OpAQ dataset. You can also use our data curation template and convert your data yourself. Accordingly, we will mention your contribution either as “Resources” or additionally as “Data curation”.

Unpublished datasets as well as data from classroom experiments are also highly appreciated.



Contribute

- **Inclusion criteria:**

- data from any studies that included high versus low anchor manipulations and estimates
- Optionally: there is a true value or unanchored mean estimate for each item

- **Incentive:** Co-authorship

- If you provide your data (even if it is already openly accessible) you are eligible for a co-authorship (CRediT: Resources)
- If you process your data so that it matches the data format (instructions and R-scripts available online), you qualify for a co-authorship (CRediT: Data curation)

Individual Differences References

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